Saturated fusion systems on groups of order 3^8

a running report

October 17, 2026

Abstract

We search for the saturated fusion systems $\mathcal{F}$ on the groups S of order 3^8 with $O_3(\mathcal{F}) = 1$ and $O^3(\mathcal{F}) = \mathcal{F}$, in the manner of Parker–Semeraro [1] but with the fast proto-essential search of Gautam [2] and the fusion-graph-free saturation test of the GAP package `FusionSystems` that accompanies this report. There are 1 396 077 groups of order 3^8 (Vaughan-Lee's database, available in GAP through the package `sglppow`). Section 3 reduces them to a short list by necessary conditions, Section 4 identifies the surviving groups, Section 5 contains the look-up tables of subgroups of $\mathrm{GL}_n(3)$ with a strongly 3-embedded subgroup, and Section 6 records the fusion systems found group by group. This document is updated as the computation proceeds; the current state is summarised in Section 1.

Contents

1 Status **1**

2 Conventions and the strategy **2**

3 The reduction **3**
3.1 Necessary conditions 3
3.2 Statistics 5

4 The surviving groups **5**
4.1 Sylow 3-subgroups of simple groups of order 3^8 5

5 Look-up tables: subgroups of $\mathrm{GL}_n(3)$ with a strongly 3-embedded subgroup **6**

6 Results group by group **14**
6.1 The survivors of Stage 1b and the status of Stage 2 14
6.2 The Sylow 3-subgroup of $\mathrm{PSL}_3(q)$, $3^4 \parallel q-1$, and of $\mathrm{PSU}_3(q)$, $3^4 \parallel q+1$: `SmallGroup(6561,2221)` 15
6.3 The other groups of maximal class: `SmallGroup(6561,`n`)`, $n = 2222, \ldots, 2226$ 17
6.4 `SmallGroup(6561,3016)`: two maximal subgroups as candidates, no fusion system . 18
6.5 Groups carrying no fusion system 18
6.6 Groups whose proto-essential subgroups are 3^4 with $|N_S(E) : E| = 3$ 18
6.7 Groups whose only proto-essential subgroups are 3^5 with $\mathrm{Out}_S(E) \cong C_9$ 18

1 Status

[updated as the work proceeds; last update 17 October 2026, evening]

- Stage 1a (cheap reduction of the 1 396 077 groups): a first run was stopped after 17 600 groups when the 3-group test was found to use a wrong quantity (Section 3); restarted the same day with the corrected pipeline on four cores, about 1.2 s per group, so roughly five to six days in total; restarted a second time the same evening (after 19 000 groups) when the look-up table for $\mathrm{GL}_5(3)$, $T \cong 3^2$, was found to lack two classes (Section 5). The statistics of Section 3.2 are refreshed from time to time. So far only the groups with a small library number survive (the library is ordered by the number of generators; in the second run all 520 survivors among the first 19 000 groups had library number at most 3016).
- Stage 1b (automorphism group tests): run on the 520 survivors so far; 42 groups survive (Section 6.1).
- Look-up tables for $\mathrm{GL}_n(3)$: complete for $n \leq 4$, and for $n = 5$, $T \cong 3^2$ (Section 5).
- Stage 2 (fusion systems group by group): running through the 42 survivors in order of difficulty (Section 6.1 lists the status of each group); the seven systems on `SmallGroup(6561,2221)` are identified (three of them are the fusion systems of $\mathrm{PSL}_3(163)$, $\mathrm{PSU}_3(647)$, of their extensions by an outer automorphism, and of ${}^3D_4(53)$; the other four are candidates for exotic systems), see Section 6.2; the other maximal class groups carry eight more systems, all candidates for exotic systems (Section 6.3).
- Open: the comparison with the classification of Díaz–Ruiz–Viruel for the two maximal class groups; the groups with a proto-essential subgroup 3^5_{e}, $T \cong C_9$ (no look-up table yet); the Sylow 3-subgroups of $\mathrm{Alt}(18)$ and $\mathrm{PSp}_4(9)$ (large elementary abelian subgroups, treated last).

2 Conventions and the strategy

Throughout $p = 3$, S is a group of order 3^8 and $\mathcal{F}$ is a saturated fusion system on S with $O_3(\mathcal{F}) = 1$ and $O^3(\mathcal{F}) = \mathcal{F}$. We write $\mathrm{Aut}_S(E) = N_S(E)/C_S(E)$ and $\mathrm{Out}_S(E) = \mathrm{Aut}_S(E)\mathrm{Inn}(E)/\mathrm{Inn}(E)$; for an S-centric subgroup E (that is $C_S(E) \leq E$) we have $\mathrm{Out}_S(E) \cong N_S(E)/E$.

By the Alperin–Goldschmidt fusion theorem $\mathcal{F}$ is generated by $\mathrm{Aut}_{\mathcal{F}}(S)$ and the groups $\mathrm{Aut}_{\mathcal{F}}(E)$ for the $\mathcal{F}$-essential subgroups E (the $\mathcal{F}$-centric, fully normalised subgroups with $\mathrm{Out}_{\mathcal{F}}(E)$ having a strongly 3-embedded subgroup). Following [1] we therefore proceed as follows.

1. Determine the subgroups $E \leq S$ which can be essential in some saturated fusion system on S (the *proto-essential* subgroups), up to $\mathrm{Aut}(S)$-conjugacy. Here we use Gautam's theorem (Theorem 2.1 below) instead of the subgroup lattice of S.
2. Determine the Borel subgroups, i.e. the $3'$-subgroups $K \leq \mathrm{Aut}(S)$ up to conjugacy, so that $\mathrm{Aut}_{\mathcal{F}}(S) = \mathrm{Inn}(S)K$ (recall that $\mathrm{Inn}(S)$ is a normal Sylow 3-subgroup of $\mathrm{Aut}_{\mathcal{F}}(S)$).
3. For each Borel subgroup and each proto-essential subgroup E determine the candidates for $\mathrm{Aut}_{\mathcal{F}}(E)$: the subgroups of $\mathrm{Aut}(E)$ containing $\mathrm{Inn}(E)$ with $\mathrm{Aut}_S(E)$ as a Sylow 3-subgroup, with Out having a strongly 3-embedded subgroup and compatible with the Borel subgroup.
4. For each choice of a set of essential subgroups (up to the action of $N_{\mathrm{Aut}(S)}(\mathrm{Aut}_{\mathcal{F}}(S))$) and automizers, build the fusion system, discard it unless $O^3(\mathcal{F}) = \mathcal{F}$, $O_3(\mathcal{F}) = 1$ and $\mathcal{F}$ is saturated, and finally remove isomorphic copies.

Steps 2–4 are only carried out for the groups surviving the reduction of Section 3. All computations are done with the GAP package `FusionSystems`; the scripts for this report are in the directory `order38/scripts` of the repository.

Theorem 2.1 (Gautam [2, Theorem 3.6])**.** *Let $S = L_1 > L_2 > \cdots > L_m = 1$ be a central series of S with all factors of order p which refines the lower central series. Every essential subgroup of a saturated fusion system on S is S-conjugate to a subgroup $C_S(xA/A) = \{s \in S : [s,x] \in A\}$ with $A = L_i \leq [S,S]$ and xA an element of order p of S/A.*

3 The reduction

3.1 Necessary conditions

We use the following necessary conditions on S. Conditions (R1)–(R3) involve only S, its candidate subgroups E and (for the few candidates passing the structural tests) $\mathrm{Aut}(E)$; they are applied to all 1 396 077 groups (Stage 1a). Conditions (R4)–(R6) need $\mathrm{Aut}(S)$ and are applied to the survivors (Stage 1b).

Lemma 3.1 ([1, Lemma 7.1])**.** (R1) *If $|S : \mathrm{Z}(S)| \leq 9$ then S carries no saturated fusion system with $O_3(\mathcal{F}) = 1$.*

Lemma 3.2. (R2) *Let E be an essential subgroup of a saturated fusion system $\mathcal{F}$ on S and $T = \mathrm{Out}_S(E) \cong N_S(E)/E$. Then*

(a) E is S-centric and $E < N_S(E)$;

(b) T is a Sylow 3-subgroup of a group with a strongly 3-embedded subgroup, so T is cyclic, or elementary abelian of rank at least 2, or $T \cong 3^{1+2}_+$, or $T \cong \mathrm{Syl}_3(\mathrm{PSU}_3(9))$ of order 3^6;

(c) $|E/\Phi(E)| \geq |T|^2$;

(d) if $1 = D_0 < D_1 < \cdots < D_k = E$ is a chain of characteristic subgroups of E, then no element of $N_S(E) \setminus E$ centralises every factor D_i/D_{i-1};

(e) $\mathrm{Aut}(E)$ is not a 3-group, $\mathrm{Aut}_S(E) \cap O_3(\mathrm{Aut}(E)) = \mathrm{Inn}(E)$, and $\mathrm{Aut}(E)$ is insoluble unless T is cyclic;

(f) if T is not cyclic then $|\mathrm{Aut}(E)|$ is divisible by 5 for $T \cong 3^2$, by $7 \cdot 13$ for $T \cong 3^3$, by 7 for $T \cong 3^{1+2}_+$, by $5 \cdot 41$ for $T \cong 3^4$, by $11 \cdot 61$ for $T \cong 3^5$ and by $5 \cdot 73$ for $|T| = 3^6$.

Proof. (a) and (b) are part of the definition and the classification of the groups with a strongly p-embedded subgroup (for $p = 3$ and T non-cyclic the possibilities for $O^{3'}(\mathrm{Out}_{\mathcal{F}}(E)/O_{3'}(\mathrm{Out}_{\mathcal{F}}(E)))$ are $\mathrm{PSL}_2(3^n)$, $\mathrm{PSU}_3(3^n)$, ${}^2G_2(3^n)$, $\mathrm{PSL}_3(4)$ and M_{11}; the Ree groups have Sylow 3-subgroups of order at least 3^9). (c) is [1, Lemma 2.?]: $\mathrm{Out}_{\mathcal{F}}(E)$ acts faithfully on $E/\Phi(E)$ because the kernel of $\mathrm{Aut}(E) \to \mathrm{Aut}(E/\Phi(E))$ is a normal 3-subgroup and $O_3(\mathrm{Out}_{\mathcal{F}}(E)) = 1$ as E is $\mathcal{F}$-radical. (d) The automorphisms of E centralising all factors of a chain of characteristic subgroups form a normal subgroup K of $\mathrm{Aut}(E)$, and K is a 3-group (a $3'$-automorphism centralising all factors of a normal series of a 3-group is trivial). Hence $K\mathrm{Inn}(E)/\mathrm{Inn}(E) \leq O_3(\mathrm{Out}(E))$ meets $\mathrm{Out}_{\mathcal{F}}(E)$ in a normal 3-subgroup, which is trivial, so $K \cap \mathrm{Aut}_S(E) \leq \mathrm{Inn}(E)$, which is (d) as E is S-centric. (e) is the $\mathcal{F}$-radical condition $O_3(\mathrm{Out}_{\mathcal{F}}(E)) = 1$ together with the fact that $T \neq 1$ is not normal in $\mathrm{Out}_{\mathcal{F}}(E)$ (a group with a strongly 3-embedded subgroup has more than one Sylow 3-subgroup), so

that $\mathrm{Out}(E)$ is not a 3-group; if T is not cyclic then $O^{3'}(\mathrm{Out}_{\mathcal{F}}(E)/O_{3'}(\mathrm{Out}_{\mathcal{F}}(E)))$ is a non-abelian simple group by the classification quoted in (b), so $\mathrm{Aut}(E)$ is insoluble. (f) That simple group is a section of $\mathrm{Out}(E)$ and is one of $\mathrm{PSL}_2(9)$, $\mathrm{PSL}_3(4)$, M_{11} for $T \cong 3^2$, $\mathrm{PSL}_2(27)$ for 3^3, $\mathrm{PSU}_3(3)$ for 3_+^{1+2}, $\mathrm{PSL}_2(81)$ for 3^4, $\mathrm{PSL}_2(243)$ for 3^5 and $\mathrm{PSL}_2(729)$ or $\mathrm{PSU}_3(9)$ for $|T| = 3^6$; the primes listed divide the orders of all groups in the respective list. □

In Stage 1a a subgroup E is called a *candidate* if it is S-conjugate to a subgroup $C_S(xA/A)$ of Theorem 2.1 and satisfies (a)–(f) of Lemma 3.2, where in (d) we use the chain $E > \Phi(E) > 1$ and then the covering pairs of the set of characteristic subgroups generated (by one round of intersections and products) by $\mathrm{Z}(E)$, $\mathrm{Z}_2(E)$, $\Phi(E)$, E', $\gamma_3(E)$, $\mho_1(E)$, $\Omega_1(\mathrm{Z}(E))$, $\mathrm{Z}(\Phi(E))$ and $C_E(E')$. (The stability group of the set of covering pairs is contained in the stability group of any maximal chain in the set, so the argument of (d) applies.) The tests are applied in this order, so that $\mathrm{Aut}(E)$ (computed with the GAP package `autpgrp`) is only needed for the candidates surviving (a)–(d); the first part of (e) discards a good part of them, since many groups of order 3^7 have a 3-group of automorphisms. With `autpgrp` we computed the automorphism groups of all groups of order 3^k, $k \le 7$ (`data/aut3table.g`):

$\lvert E\rvert$	3^3	3^4	3^5	3^6	3^7
groups	5	15	67	504	9310
$\mathrm{Aut}(E)$ a 3-group	0	0	0	30	2119
$\mathrm{Aut}(E)$ insoluble	3	5	8	17	37

(A first version of this computation used `autpgrp`'s `glOrder`, which is 1 whenever the action of $\mathrm{Aut}(E)$ on $E/\Phi(E)$ is soluble, and so counted the groups with insoluble automorphism group; the sweep was restarted with the corrected criterion.) A candidate which is not elementary abelian must therefore be isomorphic to one of the groups with $|\mathrm{Aut}(E)|$ not a power of 3; this is tested first through a cheap isomorphism invariant (the orders of the centre, Frattini and derived subgroups, the exponent, the class, the abelian invariants, the upper and lower central series, $\mho_1$, $\Omega_1(\mathrm{Z})$ and the class sizes by element order; `data/fingerprints.g`), and $\mathrm{Aut}(E)$ is only computed for candidates with a matching fingerprint. Elementary abelian candidates $E \cong 3^n$ have $\mathrm{Aut}(E) = \mathrm{GL}_n(3)$, and for $n \le 4$ the class of $\mathrm{Aut}_S(E)$ in $\mathrm{GL}_n(3)$ must be one of the classes carrying a candidate automizer in the look-up tables of Section 5.

Lemma 3.3. (R3) *If $W \neq 1$ is a characteristic subgroup of S which is contained in and characteristic in every candidate, then $W \le O_3(\mathcal{F})$ for every saturated fusion system $\mathcal{F}$ on S.*

Proof. W is normalised by $\mathrm{Aut}_{\mathcal{F}}(S)$ and by $\mathrm{Aut}_{\mathcal{F}}(E)$ for every essential E (an essential subgroup is S-conjugate to a candidate E_0, and $W = W^s$ is characteristic in E_0^s). By Alperin–Goldschmidt W is normalised by every morphism of $\mathcal{F}$ defined on it, i.e. $W \trianglelefteq \mathcal{F}$. □

In Stage 1a "characteristic" is replaced by the sufficient condition of membership in the sets of visibly characteristic subgroups described above.

Lemma 3.4. *Let $\mathcal{E}$ be a set of representatives of the S-classes of the proto-essential subgroups of S: the candidates of Stage 1a, where a candidate whose $\mathrm{Aut}(S)$-orbit contains a subgroup which is not a candidate is discarded (all the conditions of Lemma 3.2 are $\mathrm{Aut}(S)$-invariant).*

(R4) If $\mathcal{E} = \emptyset$ then $O_3(\mathcal{F}) = S$ for every saturated $\mathcal{F}$ on S.

(R5) If a nontrivial characteristic subgroup of S is characteristic in every member of $\mathcal{E}$, then $O_3(\mathcal{F}) \neq 1$.

(R6) If $S \neq \langle [S, O^3(\mathrm{Aut}(S))],\ E : E \in \mathcal{E} \rangle$ *then* $O^3(\mathcal{F}) < \mathcal{F}$ *for every saturated* $\mathcal{F}$ *on* S.

Proof. (R4) and (R5) are as before. (R6): $O^3(\mathcal{F}) = \mathcal{F}$ if and only if the hyperfocal subgroup $\langle [P, O^3(\mathrm{Aut}_{\mathcal{F}}(P))] : P \leq S \rangle$ is S, and by Alperin–Goldschmidt it is generated by $[S, O^3(\mathrm{Aut}_{\mathcal{F}}(S))]$ and the $[E, O^3(\mathrm{Aut}_{\mathcal{F}}(E))] \leq E$ for E essential; finally $O^3(\mathrm{Aut}_{\mathcal{F}}(S)) \leq O^3(\mathrm{Aut}(S))$. □

3.2 Statistics

[preliminary; updated as the sweep proceeds] The table `tables/stage1a_stats.tex` is regenerated by `scripts/analyze.py` from the output of the sweep; at the time of writing it reads as follows (the first ids of the library are rich in survivors, as the groups are ordered by rank and class).

outcome	groups
processed	3036
rejected: no candidate essential subgroup	2238
rejected: characteristic subgroup of order 3 in all candidates	131
rejected: characteristic subgroup of order 243 in all candidates	100
rejected: characteristic subgroup of order 9 in all candidates	54
rejected: characteristic subgroup of order 27 in all candidates	53
rejected: characteristic subgroup of order 81 in all candidates	22
rejected: $\lvert S : \mathrm{Z}(S)\rvert \leq 9$	14
rejected: characteristic subgroup of order 729 in all candidates	10
errors	0
survivors	414

Stage 1b has so far rejected every survivor it was given except those listed in Section 6, mostly by the characteristic subgroup test (R5), then by the generation test (R6).

4 The surviving groups

[to be filled in]

4.1 Sylow 3-subgroups of simple groups of order 3^8

The finite simple groups G with $|G|_3 = 3^8$ and non-abelian Sylow 3-subgroups are (up to the obvious isomorphisms):

- $\mathrm{Alt}(18)$, $\mathrm{Alt}(19)$, $\mathrm{Alt}(20)$ (and the symmetric groups), with $S \cong (C_3 \wr C_3) \times (C_3 \wr C_3)$;
- $\mathrm{PSp}_4(9)$, S a Sylow 3-subgroup of $\mathrm{Sp}_4(9)$;
- $\mathrm{PSL}_3(q)$ with $3^4 \parallel q - 1$ (e.g. $q = 163$) and $\mathrm{PSU}_3(q)$ with $3^4 \parallel q + 1$ (e.g. $q = 647$), with $S \cong \big(\{(x, y, z) \in C_{81}^3 : xyz = 1\} \rtimes C_3\big)/C_3$ of maximal class;
- ${}^3D_4(q)$ with $3^3 \parallel q^2 - 1$ (e.g. $q = 53,\ 107,\ 109$), where $S \cong (C_{27} \times C_{81}) \rtimes C_3$ is again `SmallGroup(6561,2221)`: a model of S inside $C_{81} \wr C_3$ (generated by $\{\mathrm{diag}(a, b, c) : a, b, c \in C_{27},\ abc = 1\}$, by $\mathrm{diag}(\zeta, \zeta, \zeta^{-2})$ with ζ of order 81, and by the 3-cycle) is validated against ${}^3D_4(2)$ (whose Sylow 3-subgroup is `SmallGroup(81,9)`, the analogous model for $3 \parallel q+1$) and is isomorphic to `SmallGroup(6561,2221)` (`scripts/d4model.g`, `scripts/reference_3d4.g`).

The groups $\mathrm{PSL}_2(3^8)$, $\mathrm{PSL}_2(q)$ with $3^8 \,\|\, q \pm 1$ and $\mathrm{PSp}_4(q)$ with $3^4 \,\|\, q^2 - 1$ have abelian Sylow 3-subgroups, which carry no fusion system with $O_3(\mathcal{F}) = 1$. No sporadic group has $|G|_3 = 3^8$. The first three groups are constructed in `scripts/known_groups.g`; their invariants are

S	$\lvert\mathrm{Z}(S)\rvert$	class	rank	exponent	$\lvert S'\rvert$	classes
$\mathrm{Syl}_3(\mathrm{Alt}(18))$	9	3	4	9	81	289
$\mathrm{Syl}_3(\mathrm{PSp}_4(9))$	9	3	4	9	81	161
$\mathrm{Syl}_3(\mathrm{PSL}_3(163))$	3	7	2	81	729	737

All three survive Stages 1a and 1b (the library numbers of the first two are not known, as `IdGroup` is not available for order 3^8; they are treated directly). The proto-essential candidates are: for $\mathrm{Syl}_3(\mathrm{Alt}(18))$ two S-classes of non-abelian subgroups of order 3^7 (ranks 4 and 5, $|N_S(E) : E| = 3$), one elementary abelian 3^6 with $\mathrm{Out}_S(E) \cong 3^2$ and two S-classes of non-abelian subgroups of order 3^6 with $|N_S(E) : E| = 3$, $|\mathrm{Aut}(S)| = 2^5 \cdot 3^{11}$; for $\mathrm{Syl}_3(\mathrm{PSp}_4(9))$ one elementary abelian 3^6 and one non-abelian subgroup of order 3^6, both with $\mathrm{Out}_S(E) \cong 3^2$, and seven S-classes of elementary abelian 3^4 with $\mathrm{Out}_S(E) \cong 3^2$, $|\mathrm{Aut}(S)| = 2^7 \cdot 3^{12}$. The elementary abelian 3^6 with $\mathrm{Out}_S(E) \cong 3^2$ requires the automizer candidates in $\mathrm{GL}_6(3)$, for which no look-up table exists yet (the perfect subgroup method of the package is complete here since $\mathrm{Out}_S(E)$ is not cyclic); these two groups are treated at the end of Stage 2.

5 Look-up tables: subgroups of $\mathrm{GL}_n(3)$ with a strongly 3-embedded subgroup

If $E \cong 3^n$ is an elementary abelian essential subgroup then $\mathrm{Out}_{\mathcal{F}}(E) = \mathrm{Aut}_{\mathcal{F}}(E) \leq \mathrm{GL}_n(3)$ has Sylow 3-subgroup $T = \mathrm{Aut}_S(E)$ of order at most 3^{8-n} (as E is centric and $N_S(E) > E$, in fact $n \leq 7$) and a strongly 3-embedded subgroup. The following tables list, for each $\mathrm{GL}_n(3)$-class of admissible T (cyclic, elementary abelian or 3^{1+2}_+), the subgroups H with $T \in \mathrm{Syl}_3(H)$ and a strongly 3-embedded subgroup, up to conjugacy in $N_{\mathrm{GL}_n(3)}(T)$. For $n \leq 4$ the tables are complete. They are produced by `scripts/gltables.g` and stored in machine-readable form in `data/gln_table.g`.

The classes of T are distinguished in the tables by $|N_{\mathrm{GL}_n(3)}(T)|$; when the same H appears several times for one T the copies are not conjugate under $N_{\mathrm{GL}_n(3)}(T)$ (they differ in the way H acts, e.g. in the determinant or in the composition factors). The column "factors" gives the dimensions of the composition factors of H on the natural module, so that e.g. $1 + 2$ for $\mathrm{SL}_2(3) \leq \mathrm{GL}_3(3)$ means the module has a trivial submodule with irreducible 2-dimensional quotient. For $n \leq 4$ the only candidates with a non-cyclic Sylow 3-subgroup occur for two classes of $T \cong 3^2$ in $\mathrm{GL}_4(3)$: the extensions of $\mathrm{Alt}(6) \cong \Omega_4^-(3)$ and of $\mathrm{SL}_2(9) = 2.\mathrm{Alt}(6)$ by subgroups of $\mathrm{Out}(\mathrm{Alt}(6))$ and by scalars; $T \cong C_9$, 3^3, 3^4 and 3^{1+2}_+ admit no candidates in $\mathrm{GL}_4(3)$. (The Ree groups and $\mathrm{PSU}_3(9)$ need $n \geq 6$, $\mathrm{PSL}_3(4)$ and M_{11} need $n \geq 5$.)

Table 1: Subgroups H of $\mathrm{GL}_2(3)$ with a strongly 3-embedded subgroup, by Sylow 3-subgroup T (one H per $N_{\mathrm{GL}_2(3)}(T)$-class; $|N_H(T)|$, $|O_{3'}(H)|$, whether H is soluble, the dimensions of the composition factors of the natural module).

T	$\lvert N(T)\rvert$	$\lvert H\rvert$	H	$\lvert N_H(T)\rvert$	$\lvert O_{3'}(H)\rvert$	sol.	factors
$C3$	12	24	`SL(2,3)`	6	8	true	2

T	$\lvert N(T)\rvert$	$\lvert H\rvert$	H	$\lvert N_H(T)\rvert$	$\lvert O_{3'}(H)\rvert$	sol.	factors
$C3$	12	48	GL(2,3)	12	8	true	2

Table 2: Subgroups H of $\mathrm{GL}_3(3)$ with a strongly 3-embedded subgroup, by Sylow 3-subgroup T (one H per $N_{\mathrm{GL}_3(3)}(T)$-class; $\lvert N_H(T)\rvert$, $\lvert O_{3'}(H)\rvert$, whether H is soluble, the dimensions of the composition factors of the natural module).

T	$\lvert N(T)\rvert$	$\lvert H\rvert$	H	$\lvert N_H(T)\rvert$	$\lvert O_{3'}(H)\rvert$	sol.	factors
$C3$	216	24	SL(2,3)	6	8	true	1+2
$C3$	216	48	GL(2,3)	12	8	true	2+1
$C3$	216	48	C2 x SL(2,3)	12	16	true	2+1
$C3$	216	48	GL(2,3)	12	8	true	2+1
$C3$	216	96	C2 x GL(2,3)	24	16	true	1+2
$C3$	36	12	A4	3	4	true	3
$C3$	36	24	C2 x A4	6	8	true	3
$C3$	36	24	S4	6	4	true	3
$C3$	36	24	S4	6	4	true	3
$C3$	36	39	C13 : C3	3	13	true	3
$C3$	36	48	C2 x S4	12	8	true	3
$C3$	36	78	C2 x (C13 : C3)	6	26	true	3
3^2	864	none					
3^2	864	none					
3^2	108	none					
3_+^{1+2}	216	none					

Table 3: Subgroups H of $\mathrm{GL}_4(3)$ with a strongly 3-embedded subgroup, by Sylow 3-subgroup T (one H per $N_{\mathrm{GL}_4(3)}(T)$-class; $\lvert N_H(T)\rvert$, $\lvert O_{3'}(H)\rvert$, whether H is soluble, the dimensions of the composition factors of the natural module).

T	$\lvert N(T)\rvert$	$\lvert H\rvert$	H	$\lvert N$
$C3$	46656	24	SL(2,3)	
$C3$	46656	48	GL(2,3)	
$C3$	46656	48	GL(2,3)	
$C3$	46656	48	C2 x SL(2,3)	
$C3$	46656	48	C2 x SL(2,3)	
$C3$	46656	48	GL(2,3)	
$C3$	46656	96	C2 x GL(2,3)	
$C3$	46656	96	C4 x SL(2,3)	
$C3$	46656	96	Q8 : (C3 : C4)	
$C3$	46656	96	C2 x C2 x SL(2,3)	
$C3$	46656	96	C2 x GL(2,3)	

T	$\lvert N(T)\rvert$	$\lvert H\rvert$	H	$\lvert N$
$C3$	46656	96	C2 x GL(2,3)	
$C3$	46656	96	C2 x GL(2,3)	
$C3$	46656	192	SL(2,3) x Q8	
$C3$	46656	192	Q8 : D24	
$C3$	46656	192	C8 x SL(2,3)	
$C3$	46656	192	Q8 : (C3 : C8)	
$C3$	46656	192	Q8 : (C3 : Q8)	
$C3$	46656	192	C4 x GL(2,3)	
$C3$	46656	192	SL(2,3) x D8	
$C3$	46656	192	Q8 : ((C6 x C2) : C2)	
$C3$	46656	192	C2 x C2 x GL(2,3)	
$C3$	46656	384	GL(2,3) x Q8	
$C3$	46656	384	SL(2,3) x QD16	
$C3$	46656	384	GL(2,3) x D8	
$C3$	46656	384	Q8 : ((C3 : Q8) : C2)	
$C3$	46656	384	C8 : GL(2,3)	
$C3$	46656	384	C8 x GL(2,3)	
$C3$	46656	384	Q8 : GL(2,3)	
$C3$	46656	768	GL(2,3) x QD16	
$C3$	648	12	A4	
$C3$	648	24	SL(2,3)	
$C3$	648	24	S4	
$C3$	648	24	S4	
$C3$	648	24	C2 x A4	
$C3$	648	24	C2 x A4	
$C3$	648	24	S4	
$C3$	648	24	C2 x A4	
$C3$	648	24	S4	
$C3$	648	39	C13 : C3	
$C3$	648	48	C2 x S4	
$C3$	648	48	C2 x C2 x A4	
$C3$	648	48	C2 x S4	
$C3$	648	48	C2 x S4	
$C3$	648	48	C2 x S4	
$C3$	648	48	C2 x S4	
$C3$	648	48	C2 x S4	
$C3$	648	48	GL(2,3)	
$C3$	648	60	A5	
$C3$	648	78	C2 x (C13 : C3)	
$C3$	648	78	C2 x (C13 : C3)	
$C3$	648	78	C2 x (C13 : C3)	
$C3$	648	96	((C2 x C2 x C2) : (C2 x C2)) : C3	
$C3$	648	96	C2 x C2 x S4	
$C3$	648	120	S5	
$C3$	648	120	C2 x A5	
$C3$	648	120	S5	

T	$\lvert N(T)\rvert$	$\lvert H\rvert$	H	$\lvert N$
$C3$	648	156	C2 x C2 x (C13 : C3)	
$C3$	648	192	(C2 x C2 x C2) : S4	
$C3$	648	192	Q8 : S4	
$C3$	648	192	((C4 x C2) : D8) : C3	
$C3$	648	240	C2 x S5	
$C3$	648	384	(C2 x C2 x C2 x C2) : S4	
$C3$	7776	24	SL(2,3)	
$C3$	7776	24	SL(2,3)	
$C3$	7776	48	GL(2,3)	
$C3$	7776	48	C2 . S4 = SL(2,3) . C2	
$C3$	7776	48	((C4 x C2) : C2) : C3	
$C3$	7776	48	C2 x SL(2,3)	
$C3$	7776	48	C2 x SL(2,3)	
$C3$	7776	48	GL(2,3)	
$C3$	7776	48	GL(2,3)	
$C3$	7776	48	GL(2,3)	
$C3$	7776	96	((C2 x C2 x C2) : (C2 x C2)) : C3	
$C3$	7776	96	((C8 x C2) : C2) : C3	
$C3$	7776	96	SL(2,3) : C4	
$C3$	7776	96	(C2 . S4 = SL(2,3) . C2) : C2	
$C3$	7776	96	((C2 x Q8) : C2) : C3	
$C3$	7776	96	((C4 x C2) : C2) : S3	
$C3$	7776	96	((C4 x C2) : C2) : S3	
$C3$	7776	96	C2 x GL(2,3)	
$C3$	7776	96	(C2 x Q8) : S3	
$C3$	7776	96	C2 x GL(2,3)	
$C3$	7776	120	SL(2,5)	
$C3$	7776	192	(Q8 x Q8) : C3	
$C3$	7776	192	((C2 x C2 x C2) : (C2 x C2)) : S3	
$C3$	7776	192	((C4 x C2) : C2) : D12	
$C3$	7776	192	((C2 . S4 = SL(2,3) . C2) : C2) : C2	
$C3$	7776	192	((C2 . S4 = SL(2,3) . C2) : C2) : C2	
$C3$	7776	192	(C2 . S4 = SL(2,3) . C2) : C4	
$C3$	7776	192	SL(2,3) : D8	
$C3$	7776	192	((C8 : C2) : (C2 x C2)) : C3	
$C3$	7776	240	SL(2,5) : C2	
$C3$	7776	240	C2 . S5 = SL(2,5) . C2	
$C3$	7776	240	SL(2,5) : C2	
$C3$	7776	384	(Q8 x Q8) : S3	
$C3$	7776	384	Q8 : GL(2,3)	
$C3$	7776	384	((C2 . ((C2 x C2 x C2) : C4) = (C4 x C2) . (C4 x C2)) : C2) : C3	
$C3$	7776	384	((C8 x C2) : C2) : D12	
$C3$	7776	480	C8 . A5 = SL(2,5) . C4	
$C3$	7776	480	SL(2,5) : (C2 x C2)	
$C3$	7776	480	(C2 . S5 = SL(2,5) . C2) : C2	
$C3$	7776	768	(Q8 x Q8) : D12	

T	$\lvert N(T)\rvert$	$\lvert H\rvert$	H	$\lvert N$
$C3$	7776	960	(C8 . A5 = SL(2,5) . C4) : C2	
$C3$	7776	1920	C2 . ((C2 x C2 x C2 x C2) : A5)	
$C3$	7776	3840	(C2 . ((C2 x C2 x C2 x C2) : A5)) : C2	
3^2	432	none		
3^2	46656	none		
3^2	648	none		
3^2	648	none		
3^2	1944	none		
3^2	46656	none		
3^2	5832	none		
$C9$	324	none		
3^2	10368	720	SL(2,9)	
3^2	10368	1440	SL(2,9) : C2	
3^2	10368	1440	SL(2,9) : C2	
3^2	10368	2880	SL(2,9) : C4	
3^2	10368	2880	SL(2,9) : (C2 x C2)	
3^2	10368	2880	SL(2,9) : C4	
3^2	10368	5760	SL(2,9) : D8	
3^2	10368	5760	SL(2,9) : Q8	
3^2	10368	5760	GL(2,9)	
3^2	10368	11520	SL(2,9) : QD16	
3^2	2592	none		
3^2	864	360	A6	
3^2	864	720	C2 x A6	
3^2	864	720	A6 . C2	
3^2	864	720	S6	
3^2	864	1440	A6 : C4	
3^2	864	1440	C2 x S6	
3^2	864	1440	C2 x (A6 . C2)	
3^2	864	2880	(C2 x (A6 . C2)) : C2	
3^3	5832	none		
3_+^{1+2}	3888	none		
3_+^{1+2}	2916	none		
3^3	7776	none		
3_+^{1+2}	1944	none		
3^3	3888	none		
3_+^{1+2}	3888	none		
3^3	7776	none		
3^3	606528	none		
3_+^{1+2}	23328	none		
3_+^{1+2}	972	none		
3^3	5832	none		
3_+^{1+2}	1944	none		
3^3	11664	none		
3_+^{1+2}	1944	none		
3_+^{1+2}	972	none		

T	$\lvert N(T)\rvert$	$\lvert H\rvert$	H	$\lvert N$
3^3	606528	none		
3_+^{1+2}	1944	none		
3^4	186624	none		

$\mathrm{GL}_5(3)$, $T \cong 3^2$. For $n = 5$ the subgroup lattice is out of reach, but for $T \cong 3^2$ the candidates can be constructed directly: $O^{3'}(H/O_{3'}(H))$ is Alt(6), $\mathrm{PSL}_3(4)$ or M_{11}, and $7 \nmid |\mathrm{GL}_5(3)|$ excludes $\mathrm{PSL}_3(4)$; hence the perfect residuum Y of H is Alt(6), $\mathrm{SL}_2(9)$ or M_{11} (M_{11} has trivial Schur multiplier), and H lies between Y and $N_{\mathrm{GL}_5(3)}(Y)$ by the Frattini argument. The faithful $\mathbb{F}_3$-modules of dimension 5 are: for $\mathrm{SL}_2(9)$ only $4+1$ (the natural module over $\mathbb{F}_9$ and a trivial module; an extension of the two splits since the centre acts differently on them), for Alt(6) the split module $4+1$ and the two non-split uniserial modules 4/1 and 1/4 (the permutation module on 6 points is uniserial 1/4/1 over $\mathbb{F}_3$ and $\mathrm{Ext}^1_{\mathrm{Alt}(6)}(\mathbb{F}_3, 4)$ is one-dimensional, so the augmentation submodule and its dual are the only non-split extensions), and for M_{11} its two dual irreducible 5-dimensional modules (constituents of the module induced from a 5-dimensional constituent of the permutation module of $\mathrm{PSL}_2(11) < M_{11}$ on 11 points). The Sylow 3-subgroups of the non-split embeddings of Alt(6) are not conjugate to that of $4+1$, so they give two further classes of T. The script is `scripts/gl5direct.g`; the six classes of $T \cong 3^2$ in $\mathrm{GL}_5(3)$ with candidates carry 34, 27, 2, 2, 5 and 5 of them. [The first version of this table omitted the two non-split embeddings of Alt(6); the Stage 1a sweep, which uses the table to discard candidates, was restarted a second time on 17 October 2026 (evening) with the corrected table.]

Table 4: Subgroups H of $\mathrm{GL}_5(3)$ with a strongly 3-embedded subgroup, by Sylow 3-subgroup T (one H per $N_{\mathrm{GL}_5(3)}(T)$-class; $|N_H(T)|$, $|O_{3'}(H)|$, whether H is soluble, the dimensions of the composition factors of the natural module).

T	$\lvert N(T)\rvert$	$\lvert H\rvert$	H	$\lvert N_H(T)\rvert$	$\lvert O_{3'}(H)\rvert$	sol.	factors
3^2	1679616	720	`SL(2,9)`	72	2	false	1+4
3^2	1679616	23040	`C2 x (SL(2,9) : QD16)`	2304	16	false	1+4
3^2	1679616	1440	`SL(2,9) : C2`	144	2	false	4+1
3^2	1679616	1440	`SL(2,9) : C2`	144	2	false	1+4
3^2	1679616	1440	`SL(2,9) : C2`	144	4	false	1+4
3^2	1679616	1440	`C2 x SL(2,9)`	144	4	false	4+1
3^2	1679616	1440	`SL(2,9) : C2`	144	4	false	4+1
3^2	1679616	2880	`SL(2,9) : (C2 x C2)`	288	4	false	4+1
3^2	1679616	2880	`SL(2,9) : C4`	288	8	false	1+4
3^2	1679616	2880	`SL(2,9) : C4`	288	4	false	1+4
3^2	1679616	2880	`SL(2,9) : (C2 x C2)`	288	4	false	1+4
3^2	1679616	2880	`SL(2,9) : C4`	288	8	false	4+1
3^2	1679616	2880	`C2 x (SL(2,9) : C2)`	288	4	false	1+4
3^2	1679616	2880	`SL(2,9) : (C2 x C2)`	288	4	false	4+1
3^2	1679616	2880	`C2 x (SL(2,9) : C2)`	288	8	false	1+4
3^2	1679616	2880	`SL(2,9) : C4`	288	4	false	1+4
3^2	1679616	5760	`SL(2,9) : Q8`	576	8	false	1+4

T	$\lvert N(T)\rvert$	$\lvert H\rvert$	H	$\lvert N_H(T)\rvert$	$\lvert O_{3'}(H)\rvert$	sol.	factors
3^2	1679616	5760	C2 x (SL(2,9) : C4)	576	8	false	4+1
3^2	1679616	5760	C2 x (SL(2,9) : C4)	576	16	false	4+1
3^2	1679616	5760	SL(2,9) : Q8	576	8	false	4+1
3^2	1679616	5760	C2 x (SL(2,9) : (C2 x C2))	576	8	false	1+4
3^2	1679616	5760	SL(2,9) : D8	576	8	false	1+4
3^2	1679616	5760	SL(2,9) : D8	576	8	false	4+1
3^2	1679616	5760	GL(2,9)	576	8	false	1+4
3^2	1679616	5760	GL(2,9)	576	8	false	4+1
3^2	1679616	5760	SL(2,9) : Q8	576	8	false	4+1
3^2	1679616	5760	SL(2,9) : D8	576	8	false	1+4
3^2	1679616	11520	C2 x (SL(2,9) : Q8)	1152	16	false	4+1
3^2	1679616	11520	SL(2,9) : QD16	1152	8	false	4+1
3^2	1679616	11520	C2 x (SL(2,9) : D8)	1152	16	false	4+1
3^2	1679616	11520	SL(2,9) : QD16	1152	8	false	4+1
3^2	1679616	11520	C2 x GL(2,9)	1152	16	false	1+4
3^2	1679616	11520	SL(2,9) : QD16	1152	8	false	4+1
3^2	1679616	11520	SL(2,9) : QD16	1152	8	false	4+1
3^2	15552	360	A6	36	1	false	4+1
3^2	15552	5760	C2 x ((C2 x (A6 . C2)) : C2)	576	4	false	4+1
3^2	15552	720	S6	72	1	false	1+4
3^2	15552	720	C2 x A6	72	2	false	1+4
3^2	15552	720	A6 . C2	72	1	false	4+1
3^2	15552	720	A6 . C2	72	1	false	4+1
3^2	15552	720	C2 x A6	72	2	false	1+4
3^2	15552	720	S6	72	1	false	4+1
3^2	15552	720	C2 x A6	72	2	false	4+1
3^2	15552	1440	C2 x (A6 . C2)	144	2	false	4+1
3^2	15552	1440	C2 x (A6 . C2)	144	2	false	1+4
3^2	15552	1440	C2 x S6	144	2	false	1+4
3^2	15552	1440	C2 x S6	144	2	false	4+1
3^2	15552	1440	C2 x C2 x A6	144	4	false	1+4
3^2	15552	1440	C2 x S6	144	2	false	4+1
3^2	15552	1440	C2 x (A6 . C2)	144	2	false	1+4
3^2	15552	1440	A6 : C4	144	2	false	4+1
3^2	15552	1440	C2 x S6	144	2	false	1+4
3^2	15552	1440	C2 x (A6 . C2)	144	2	false	1+4
3^2	15552	1440	A6 : C4	144	2	false	4+1
3^2	15552	2880	C2 x (A6 : C4)	288	4	false	1+4
3^2	15552	2880	(A6 : C4) : C2	288	2	false	4+1
3^2	15552	2880	(C2 x (A6 . C2)) : C2	288	2	false	4+1
3^2	15552	2880	C2 x C2 x S6	288	4	false	4+1
3^2	15552	2880	C2 x C2 x (A6 . C2)	288	4	false	1+4
3^2	15552	2880	(A6 : C4) : C2	288	2	false	1+4
3^2	15552	2880	(A6 : C4) : C2	288	2	false	4+1
3^2	2592	7920	M11	144	1	false	5
3^2	2592	15840	C2 x M11	288	2	false	5

T	$\lvert N(T)\rvert$	$\lvert H\rvert$	H	$\lvert N_H(T)\rvert$	$\lvert O_{3'}(H)\rvert$	sol.	factors
3^2	2592	7920	`M11`	144	1	false	5
3^2	2592	15840	`C2 x M11`	288	2	false	5
3^2	1296	360	`A6`	36	1	false	1+4
3^2	1296	1440	`C2 x S6`	144	2	false	1+4
3^2	1296	720	`C2 x A6`	72	2	false	1+4
3^2	1296	720	`S6`	72	1	false	1+4
3^2	1296	720	`S6`	72	1	false	1+4
3^2	1296	360	`A6`	36	1	false	4+1
3^2	1296	1440	`C2 x S6`	144	2	false	4+1
3^2	1296	720	`C2 x A6`	72	2	false	4+1
3^2	1296	720	`S6`	72	1	false	4+1
3^2	1296	720	`S6`	72	1	false	4+1

Completeness of the automizer candidates. For an essential subgroup E the candidates for $\mathrm{Out}_{\mathcal{F}}(E)$ are complete in the following cases: $|\mathrm{Out}(E)| \le 5 \cdot 10^5$ (or $\mathrm{Out}(E)$ soluble of order at most 10^9), where the subgroup lattice of $\mathrm{Out}(E)$ is used; E elementary abelian of rank at most 4 (the tables above); E elementary abelian of rank 5 with $\mathrm{Out}_S(E) \cong 3^2$ (the table above); and every E with $\mathrm{Out}_S(E)$ neither cyclic nor quaternion (then $\mathrm{Out}_{\mathcal{F}}(E)$ is not 3-soluble and the perfect subgroup method of the package is complete). For cyclic $\mathrm{Out}_S(E) = T$ and $\mathrm{Out}(E)$ beyond the lattice bound the 3-soluble candidates are now enumerated completely as well (`FS_PSolubleCyclicCandidates`): such a candidate H has $Q = O_{3'}(H) \neq 1$ with T acting faithfully on Q, and H/Q has a normal Sylow 3-subgroup, so $H = QT'R$ with T' a conjugate of T normalising Q and R a $3'$-subgroup of $N(QT')$. The $3'$-subgroups Q of $G = \mathrm{Out}(E)$ are found through the soluble radical (the $3'$-subgroups of $G/\mathrm{Rad}(G)$ from its subgroup lattice, then the Hall $3'$-subgroups of their soluble preimages, which contain every $3'$-subgroup), the conjugates T' from the conjugacy classes of $N_G(Q)$ when this group is small and as the fixed points of T on the cosets of $N_G(Q)$ otherwise. The method applies whenever $G/\mathrm{Rad}(G)$ has a feasible subgroup lattice, which is always the case for the non-abelian proto-essential subgroups met so far (where $G/\mathrm{Rad}(G)$ is a subgroup of $\mathrm{GL}_d(3)$, $d \le 5$, such as $\mathrm{PSL}_3(3)$); a soluble $\mathrm{Out}(E)$ with non-cyclic $\mathrm{Out}_S(E)$ has no candidates at all. The only remaining incomplete case is an elementary abelian E of rank 5, 6 or 7 with $\mathrm{Out}_S(E)$ cyclic and $\mathrm{Out}(E) = \mathrm{GL}_n(3)$ (where $G/\mathrm{Rad}(G)$ is $\mathrm{PGL}_n(3)$ itself), where the 3-soluble candidates are only found inside the monomial groups of the T-invariant frames; the package records every incomplete enumeration in `FS_Diagnostics`, and the Stage 2 logs and tables mark the affected groups. Results for groups whose only proto-essential subgroups of this kind are elementary abelian of rank at least 5 with cyclic $\mathrm{Out}_S(E)$ are therefore marked "complete up to the soluble candidates" below; such groups are, by the ordering of Stage 2, treated last. (For $\mathrm{Out}_S(E) \cong C_9$ the subgroup $\Omega_1(T)$ must act nontrivially on $O_{3'}(H)$, so T acts faithfully on a $\{2,5,11,13\}$-group inside $\mathrm{GL}_n(3)$; a faithful action of C_9 needs a 2-section of rank at least 6, a 5- or 11-section of rank at least 6 or a 13-section of rank at least 3, none of which exists in $\mathrm{GL}_5(3)$. So for $n = 5$ and cyclic $\mathrm{Out}_S(E)$ of order 9 there are no 3-soluble candidates; there are no others either, since a non-3-soluble candidate would contain a non-abelian simple section with cyclic Sylow 3-subgroups of order 9 and order dividing $|\mathrm{GL}_5(3)| = 2^{10} \cdot 3^{10} \cdot 5 \cdot 11^2 \cdot 13$, i.e. $\mathrm{PSL}_2(8)$, whose smallest faithful $\mathbb{F}_3$-module has dimension 7. Hence for $n = 5$, $\mathrm{Out}_S(E) \cong C_9$ there is no candidate at all, and the incompleteness only concerns $\mathrm{Out}_S(E) \cong C_3$ for $n = 5$ and cyclic $\mathrm{Out}_S(E)$ for $n = 6, 7$.)

Use of the tables. In Stage 1a an elementary abelian candidate $E \cong 3^n$ is discarded unless $\mathrm{Aut}_S(E)$ is conjugate in $\mathrm{GL}_n(3)$ to a class with candidates ($n \leq 4$, and $n = 5$ for $\mathrm{Out}_S(E) \cong 3^2$), and in Stage 2 the automizer candidates for such E are taken from the tables (transported along the conjugating matrix) instead of being enumerated anew for each group. This matters for correctness as well as for speed: for $|\mathrm{GL}_n(3)|$ beyond the lattice bound the package's own enumeration of the soluble candidates for cyclic $\mathrm{Out}_S(E)$ (the frame method) is not complete, e.g. it misses the irreducible $13{:}3 \leq \mathrm{GL}_3(3) \leq \mathrm{GL}_4(3)$.

6 Results group by group

Each subsection treats one group S and lists the saturated fusion systems $\mathcal{F}$ on S with $O_3(\mathcal{F}) = 1$ and $O^3(\mathcal{F}) = \mathcal{F}$ found by `AllFusionSystems`, up to isomorphism. For each $\mathcal{F}$ we give $\mathrm{Out}_{\mathcal{F}}(S)$, the essential subgroups E up to $\mathcal{F}$-conjugacy (with $|N_S(E) : E|$) and $\mathrm{Out}_{\mathcal{F}}(E)$, in the style of the tables of [1]. Identifications with fusion systems of finite groups are proved by `IsIsomorphicFusionSystems` against the fusion system of the group when the group is available in GAP, and are otherwise indicated as expectations. The groups are treated in the order of difficulty (the order of the largest elementary abelian proto-essential subgroup).

6.1 The survivors of Stage 1b and the status of Stage 2

Stage 1b (the tests R3, R5 and R6 of Section 3, which need $\mathrm{Aut}(S)$) was applied to the survivors of Stage 1a found so far:

outcome of Stage 1b	groups
processed	520
reject: characteristic subgroup in all proto-essentials	460
reject: [S,O^3(Aut(S))] and the proto-essentials do not generate S	18
survivors	42

The 42 survivors are listed below, in the order in which Stage 2 treats them (increasing order of the largest elementary abelian proto-essential subgroup, the subscript e marks elementary abelian candidates, a number in parentheses is $|N_S(E) : E|$ when it is not 3, and $\times k$ counts S-classes); $d(S)$ is the number of generators and $|\mathrm{Aut}(S)|_{3'}$ the $3'$-part of the order of the automorphism group. The column "Stage 2" gives the number of saturated fusion systems with $O_3(\mathcal{F}) = 1 = O^3(\mathcal{F})$ found by `AllFusionSystems`.

n	$\lvert Z(S)\rvert$	class	$d(S)$	$\exp S$	$\lvert \mathrm{Aut}(S)\rvert_{3'}$	proto-essential candidates	Stage 2	
567	9	5	2	27	8	3^5, $3^4{\times}3$	0 systems	
956	27	5	2	27	4	3^7, 3^5, 3^4	0 systems	
968	9	5	2	27	2	3^7, 3^5, 3^4	0 systems	
1701	9	5	2	27	1	3^7, 3^5, $3^4{\times}3$	0 systems	
2004	3	6	2	27	2	3^7, $3^4{\times}9$	0 systems	
2195	9	6	2	27	2	3^7, 3^4	0 systems	
3016	27	3	3	9	8	$3^7{\times}2$	0 systems	
2221	3	7	2	81	4	$3^3{\times}3$, $3^2_e{\times}3$	7 systems	Sylow of $\mathrm{PSL}_3(163)$, $\mathrm{PSU}_3(64$
2223	3	7	2	81	4	3^3, 3^2_e	2 systems	
2224	3	7	2	81	4	3^3, 3^2_e	2 systems	
2225	3	7	2	81	2	3^3, 3^2_e	1 system	Section 6.3
2226	3	7	2	81	2	$3^3{\times}4$, $3^2_e{\times}4$	3 systems	
1988	9	6	2	27	4	3^7, $3^4{\times}9$, $3^3_e{\times}9$	running	
1989	9	6	2	27	1	3^7, $3^4{\times}5$, $3^3_e{\times}5$	0 systems	
1992	3	6	2	27	4	$3^4{\times}9$, $3^3_e{\times}9$	0 systems	
1996	3	6	2	27	4	$3^4{\times}9$, $3^3_e{\times}9$	0 systems	
2006	3	6	2	27	1	$3^4{\times}10$, $3^3_e{\times}5$	0 systems	
2009	3	6	2	27	1	$3^4{\times}10$, $3^3_e{\times}5$	0 systems	
2012	3	6	2	27	1	$3^4{\times}10$, $3^3_e{\times}5$	0 systems	
2015	3	6	2	27	2	3^4, $3^3_e{\times}3$	0 systems	
2018	3	6	2	27	2	3^4, $3^3_e{\times}3$	0 systems	
2021	3	6	2	27	2	3^7, 3^4, $3^3_e{\times}3$	0 systems	Section 6.5
2024	9	6	2	27	4	3^7, 3^4, 3^3_e	0 systems	
2025	9	6	2	27	2	3^7, 3^4, 3^3_e	0 systems	Section 6.5
2029	3	6	2	27	2	3^4, 3^3_e	0 systems	Section 6.5
2030	3	6	2	27	4	3^4, 3^3_e	0 systems	
2034	3	6	2	27	2	3^4, 3^3_e	0 systems	
2035	3	6	2	27	4	3^4, 3^3_e	0 systems	
2193	9	6	2	27	4	3^7, 3^4, 3^3_e	pending	Section 6.5
2205	3	6	2	27	2	3^4, $3^3_e{\times}3$	0 systems	Section 6.5
1111	9	5	2	27	4	3^5, $3^4_e{\times}3$	pending	Section 6.6
1115	9	5	2	27	2	3^5, $3^4_e{\times}3$	pending	Section 6.6
1167	27	5	2	27	4	$3^5{\times}3$, $3^4_e{\times}5$	pending	Section 6.6
1176	9	5	2	27	2	$3^5{\times}3$, $3^4_e{\times}9$	pending	Section 6.6
1225	9	5	2	27	4	$3^5{\times}3$, $3^4_e{\times}9$	pending	Section 6.6
1229	9	5	2	27	2	$3^5{\times}3$, $3^4_e{\times}9$	pending	Section 6.6
1044	9	5	2	9	2	$3^5_e(9){\times}3$	pending	
1370	9	5	2	9	4	$3^5_e(9)$	pending	
1376	9	5	2	9	4	$3^5_e(9){\times}3$	pending	
1380	9	5	2	9	2	$3^5_e(9){\times}3$	pending	
1570	9	5	2	9	4	$3^5_e(9)$	pending	
1576	9	5	2	9	4	$3^5_e(9){\times}3$	pending	

6.2 The Sylow 3-subgroup of $\mathrm{PSL}_3(q)$, $3^4 \parallel q{-}1$, and of $\mathrm{PSU}_3(q)$, $3^4 \parallel q{+}1$: `SmallGroup(6561,2221)`

Let $T_3 = \{\mathrm{diag}(a,b,c) : a,b,c \in C_{81},\ abc = 1\} \cong C_{81}^2$, let c be the permutation matrix of a 3-cycle and $Z = \langle \omega I \rangle$ (ω a primitive cube root of unity). Then $S = (T_3 \rtimes \langle c \rangle)/Z$ is `SmallGroup(6561,2221)` (verified with `IsomorphismGroups`); it is the Sylow 3-subgroup of $\mathrm{PSL}_3(q)$ for $3^4 \parallel q{-}1$ (the smallest case is $q = 163$) and of $\mathrm{PSU}_3(q)$ for $3^4 \parallel q+1$ (smallest case $q = 647$). It has maximal class, $|\mathrm{Z}(S)| = 3$, exponent 81, the abelian maximal subgroup $A = T_3/Z \cong C_{27} \times C_{81}$, all 4374 elements of $S \setminus A$ have order 3, and $|\mathrm{Aut}(S)| = 2^2 \cdot 3^{13}$. Let z generate $\mathrm{Z}(S)$ (the image of $d = \mathrm{diag}(1, \omega, \omega^2)$).

Proto-essential subgroups. An S-centric elementary abelian subgroup of order 9 contains z, hence is $V = \langle z, x\rangle$ with $x \in S \setminus A$ of order 3. There are exactly three S-classes V_1, V_2, V_3 of such subgroups (one for each A-class of elements of the coset cA; note $z \in [A, S]$), each with $|N_S(V_i) : V_i| = 3$, and $\mathrm{Aut}(S)$ permutes them transitively. By the $\mathrm{GL}_2(3)$ table, $\mathrm{Out}_{\mathcal{F}}(V_i) \in \{\mathrm{SL}_2(3), \mathrm{GL}_2(3)\}$. The remaining candidates are three S-classes (one $\mathrm{Aut}(S)$-class) of subgroups $E \cong 3^{1+2}_+$ with $|N_S(E) : E| = 3$. There are five Borel subgroups, $|\mathrm{Out}_{\mathcal{F}}(S)| \in \{1, 2, 2, 2, 4\}$; the Borel with $\mathrm{Out}_{\mathcal{F}}(S) \cong C_2^2$ fixes one of the three classes of rank two subgroups, which we call V_1, and interchanges V_2 and V_3 (so V_2 and V_3 are $\mathcal{F}$-conjugate whenever $\mathrm{Out}_{\mathcal{F}}(S) \cong C_2^2$). The computation (`AllFusionSystems`, 17 seconds) gives exactly 7 fusion systems. In the table the essential subgroups are listed up to $\mathcal{F}$-conjugacy; [2] marks an $\mathcal{F}$-class consisting of two S-classes.

$\mathcal{F}$	$\mathrm{Out}_{\mathcal{F}}(S)$	essential subgroups E (all with $\lvert N_S(E) : E\rvert = 3$)	$\mathrm{Out}_{\mathcal{F}}(E)$	realised by
1	C_2	V_1, V_2, V_3	$\mathrm{SL}_2(3)$ each	$\mathrm{PSL}_3(163)$, $\mathrm{PSU}_3(647)$
2	C_2	two of the V_i	$\mathrm{SL}_2(3)$ each	none known (exotic?)
3	C_2	one V_i	$\mathrm{SL}_2(3)$	none known (exotic?)
4	C_2^2	$E \cong 3^{1+2}_+$; $V_2 \sim V_3$ [2]	$\mathrm{GL}_2(3)$; $\mathrm{SL}_2(3)$	${}^3D_4(53)$ (see text)
5	C_2^2	V_1; $V_2 \sim V_3$ [2]	$\mathrm{GL}_2(3)$; $\mathrm{SL}_2(3)$	$\mathrm{PSL}_3(163)$:2, $\mathrm{PSU}_3(647)$:2
6	C_2^2	V_1	$\mathrm{GL}_2(3)$	none known (exotic?)
7	C_2^2	$V_2 \sim V_3$ [2]	$\mathrm{SL}_2(3)$	none known (exotic?)

Identification. For $G = \mathrm{PSL}_3(q)$ every element of order 3 in $S \setminus A$ has eigenvalues $1, \omega, \omega^2$ and is therefore G-conjugate to z; the preimage of each V_i in $\mathrm{SL}_3(q)$ is the Heisenberg group 3^{1+2}_+ and $N_G(V_i) \cong 3^2{:}\mathrm{SL}_2(3)$ (the Hessian group, which lies in $\mathrm{PSL}_3(q)$ exactly when $9 \mid q-1$), so all three V_i are essential in $\mathcal{F}_S(G)$ with $\mathrm{Out}_G(V_i) \cong \mathrm{SL}_2(3)$, $\mathrm{Out}_G(S) \cong C_2$ (the transposition of the Weyl group), and no 3^{1+2}_+ is essential. The graph automorphism normalises one of the three classes, where it enlarges the automiser to $\mathrm{GL}_2(3)$, and interchanges the other two. The unitary case is identical with $q+1$ in place of $q-1$. These descriptions were confirmed by computing $\mathcal{F}_S(G)$ for $G = \mathrm{PSL}_3(163)$, $\mathrm{PSL}_3(163)$:2 (graph automorphism), $\mathrm{PSU}_3(647)$ and $\mathrm{PSU}_3(647)$:2 (field automorphism) and testing isomorphism with the seven systems (`scripts/reference_l3u3.g`, log `logs/reference_l3u3.log`; 3 to 40 seconds per group). Since $\mathrm{PSL}_3(163)$ has smallest permutation degree 26 733 and $\mathrm{PSU}_3(647)$ is far beyond any permutation representation, the groups are never constructed: for $U \le S$ with $z \in \mathrm{Z}(U)$ the normaliser $N_G(U)$ is generated by $N_{C_G(z)}(U)$, where $C_G(z) = (T \rtimes \langle c\rangle)/Z$ is the monomial group of order $3(q \mp 1)^2/3$, together with one element $g_y \in N_G(U)$ with $z^{g_y} = y$ for each y in the $N_G(U)$-orbit of z; $g_y = hg_0$ with g_0 an explicit conjugator $z \to y$ (a matrix of eigenvectors of a lift of y, normalised to determinant 1 and, in the unitary case, to orthonormal rows) and $h \in C_G(z)$ a conjugator of U to $U^{g_0^{-1}} \le C_G(z)$. The extensions by the outer automorphism are handled in 6-dimensional matrix models. The systems so obtained were checked to be saturated with $O_3(\mathcal{F}) = 1 = O^3(\mathcal{F})$ by the package before the isomorphism test. (This computation uncovered a bug in the isomorphism test of the package: two systems whose stored lists of essential subgroups contained different numbers of S-class representatives of the same $\mathcal{F}$-class were declared non-isomorphic. It is fixed, the test suite passes, and none of the results reported elsewhere in this document used the affected code path.)

The group ${}^3D_4(q)$. The Sylow 3-subgroups of ${}^3D_4(q)$ for $3^3 \parallel q^2 - 1$ are also isomorphic to S (Section 4.1). ${}^3D_4(53)$ is not available as a permutation or matrix group here, but the whole situation has an exact analogue one step down: for $3 \parallel q^2 - 1$ the Sylow 3-subgroups of ${}^3D_4(q)$, $\mathrm{PSL}_3(q')$ $(9 \parallel q' - 1)$ and $\mathrm{PSU}_3(q')$ $(9 \parallel q' + 1)$ are `SmallGroup(81,9)`, and `AllFusionSystems` finds

on this group again exactly seven systems, with the same table as above (three S-classes of $V \cong 3^2$ with $|N_S(V) : V| = 3$, the same automisers, $|\mathrm{Out}_{\mathcal{F}}(S)| \in \{2, 4\}$); there $\mathcal{F}_1 = \mathcal{F}_S(\mathrm{PSL}_3(19)) = \mathcal{F}_S(\mathrm{PSU}_3(17))$ and $\mathcal{F}_4 = \mathcal{F}_S({}^3D_4(2))$ (${}^3D_4(2)$ from the library of primitive groups of degree 819; `scripts/reference_3d4.g`, `logs/reference_3d4.log`). Since the 3-local structure of ${}^3D_4(q)$ which produces the essential subgroups (the normaliser $(q^2 + \epsilon q + 1)^2{:}\mathrm{SL}_2(3)$ of a maximal torus, the subgroup $(q^2 + \epsilon q + 1) \circ \mathrm{SL}_3^{\epsilon}(q)).3$ containing S, and the subgroups $G_2(q) \supset \mathrm{SL}_3^{\epsilon}(q).2$ which contain $3_+^{1+2}{:}\mathrm{GL}_2(3)$-type normalisers) does not depend on the 3-adic valuation of $q^2 - 1$, we identify $\mathcal{F}_4$ with $\mathcal{F}_S({}^3D_4(q))$, $3^3 \parallel q^2 - 1$. [A direct computation of $\mathcal{F}_S({}^3D_4(53))$, e.g. from an explicit construction of ${}^3D_4(53)$ in $\Omega_8^+(53^3)$ by twisted root subgroups, would make this rigorous.]

Exotic systems. To our knowledge the almost simple groups with a Sylow 3-subgroup isomorphic to S are $\mathrm{PSL}_3(q)$, $\mathrm{PSU}_3(q)$ (with $3^4 \parallel q \mp 1$), their extensions by an outer automorphism of order 2, and ${}^3D_4(q)$ with $3^3 \parallel q^2 - 1$, so that $\mathcal{F}_2$, $\mathcal{F}_3$, $\mathcal{F}_6$ and $\mathcal{F}_7$ are candidates for exotic fusion systems; the same four shapes are the unrealised ones on `SmallGroup(81,9)`. Saturated fusion systems on 3-groups of maximal class were classified by Díaz, Ruiz and Viruel (*All p-local finite groups of rank two for odd prime p*, Trans. AMS 359 (2007)), who found exotic systems on these groups; the comparison of $\mathcal{F}_2$, $\mathcal{F}_3$, $\mathcal{F}_6$, $\mathcal{F}_7$ with their table is still to be made (the paper could not be accessed from the computing environment).

6.3 The other groups of maximal class: `SmallGroup(6561,`n`)`, $n = 2222, \ldots, 2226$

The six groups of maximal class and order 3^8 are `SmallGroup(6561,`n`)`, $2221 \le n \le 2226$; all have $|\mathrm{Z}(S)| = 3$, two generators, exponent 81 and $|\mathrm{Aut}(S)| = 2^a \cdot 3^b$ with $a \le 2$. The first four have an abelian maximal subgroup $A \cong C_{27} \times C_{81}$, the last two have none. They are distinguished by the orders of the elements outside $\gamma_1(S)$ (for $n \le 2224$ this is A): all of order 3 for 2221, all of order 9 for 2222 (which therefore has no rank two S-centric subgroup outside A and fails Stage 1a), and both orders for the others. The rank two essential subgroups are again $V = \langle z, x \rangle$ with $x \notin \gamma_1(S)$ of order 3, $|N_S(V) : V| = 3$, and the other proto-essential candidates are extraspecial 3_+^{1+2} with $|N_S(E) : E| = 3$; the results of `AllFusionSystems` are (2 to 7 seconds each):

n	$\lvert\mathrm{Aut}(S)\rvert$	$\mathcal{F}$	$\mathrm{Out}_{\mathcal{F}}(S)$	essential subgroups, $\mathrm{Out}_{\mathcal{F}}(E)$	realised by
2223	$2^2 \cdot 3^{12}$	1	C_2	V, $\mathrm{SL}_2(3)$	none known
		2	C_2^2	V, $\mathrm{GL}_2(3)$	none known
2224	$2^2 \cdot 3^{12}$	1	C_2	V, $\mathrm{SL}_2(3)$	none known
		2	C_2^2	V, $\mathrm{GL}_2(3)$	none known
2225	$2 \cdot 3^{12}$	1	C_2	V, $\mathrm{SL}_2(3)$	none known
2226	$2 \cdot 3^{12}$	1	C_2	V_1, V_2 (two S-classes), $\mathrm{SL}_2(3)$ each	none known
		2	C_2	V_1, $\mathrm{SL}_2(3)$	none known
		3	C_2	V_2, $\mathrm{SL}_2(3)$	none known

(For 2226 the systems 2 and 3 are not isomorphic: the two S-classes of rank two essential subgroups are not interchanged by $\mathrm{Aut}(S)$.) None of these groups is a Sylow 3-subgroup of a finite simple group (the maximal class case of Section 4.1 is `2221` only), so all eight systems are candidates for exotic fusion systems; to be compared with the tables of Díaz–Ruiz–Viruel, which cover all 3-groups of maximal class.

6.4 `SmallGroup(6561,3016)`: two maximal subgroups as candidates, no fusion system

$S =$ `SmallGroup(6561,3016)` has $|\mathrm{Z}(S)| = 27$, class 3, three generators, exponent 9 and $|\mathrm{Aut}(S)| = 2^3 \cdot 3^{15}$. Its only proto-essential candidates are two maximal subgroups E_1, E_2 (of ranks 4 and 5) with $\mathrm{Out}_S(E_i) \cong C_3$, and $|\mathrm{Out}(E_1)| = 2^6 \cdot 3^{15} \cdot 13$ (with a section $\mathrm{PSL}_3(3)$), $|\mathrm{Out}(E_2)| = 2^6 \cdot 3^{12}$ (soluble). Neither has an automizer candidate: there is no subgroup of $\mathrm{Out}(E_i)$ with $\mathrm{Out}_S(E_i)$ as a Sylow 3-subgroup and a strongly 3-embedded subgroup (the 3-soluble candidates are enumerated completely by the method of Section 5, the others by the perfect subgroup method). Hence there is no fusion system (38 seconds; the first attempt with the subgroup lattice of $\mathrm{Out}(E_2)$ ran for more than 40 minutes).

6.5 Groups carrying no fusion system

The table of Section 6.1 lists the groups for which `AllFusionSystems` finds no saturated fusion system with $O_3(\mathcal{F}) = 1$ and $O^3(\mathcal{F}) = \mathcal{F}$ (all with complete enumerations of the automizer candidates); most of them take a few seconds. Typical examples are five 2-generator groups of class 6 and exponent 27 whose only proto-essential subgroups are $E \cong 3^3$ with $|N_S(E) : E| = 3$ (recomputed with the corrected pipeline):

S	$\lvert\mathrm{Z}(S)\rvert$	$\lvert S'\rvert$	$\lvert\mathrm{Aut}(S)\rvert$	abelian max. subgroup	proto-essential classes
`SmallGroup(6561,2025)`	9	3^6	$2 \cdot 3^{12}$	no	1
`SmallGroup(6561,2029)`	3	3^6	$2 \cdot 3^{12}$	no	1
`SmallGroup(6561,2193)`	9	3^5	$4 \cdot 3^{12}$	yes	1
`SmallGroup(6561,2021)`	3	3^6	$2 \cdot 3^{11}$	no	3
`SmallGroup(6561,2205)`	3	3^5	$2 \cdot 3^{12}$	no	3

In each case the Borel subgroup $\mathrm{Aut}_{\mathcal{F}}(S) = \mathrm{Inn}(S)$ is excluded by the focal subgroup and for $|\mathrm{Out}_{\mathcal{F}}(S)| = 2$ no choice of automizers ($\mathrm{Out}_{\mathcal{F}}(E) \leq \mathrm{GL}_3(3)$ from the table of Section 5) gives a saturated system.

6.6 Groups whose proto-essential subgroups are 3^4 with $|N_S(E) : E| = 3$

Six groups passing Stage 1b have proto-essential subgroups $E \cong 3^4$, $|N_S(E) : E| = 3$ (so $\mathrm{Out}_{\mathcal{F}}(E)$ is one of the 30, 33 or 43 candidates of the three classes of C_3 in the $\mathrm{GL}_4(3)$ table) and non-abelian candidates of order 3^5: `SmallGroup(6561,`n`)` for $n = 1111$ (3 classes of E), 1115 (3), 1167 (5), 1176 (9), 1225 (9), 1229 (9). In the first (invalidated) run none of them carried a saturated fusion system with $O_3(\mathcal{F}) = 1$ and $O^3(\mathcal{F}) = \mathcal{F}$ (44 to 143 seconds each); they are being recomputed with the corrected pipeline (see the table of Section 6.1).

6.7 Groups whose only proto-essential subgroups are 3^5 with $\mathrm{Out}_S(E) \cong C_9$

Six groups passing Stage 1b (`SmallGroup(6561,`n`)` for $n = 1044$, 1370, 1376, 1380, 1570, 1576) have as proto-essential subgroups only elementary abelian $E \cong 3^5$ with $\mathrm{Out}_S(E) \cong C_9$ (one or three S-classes). There is no subgroup of $\mathrm{GL}_5(3)$ with a cyclic Sylow 3-subgroup of order 9 and a strongly 3-embedded subgroup (Section 5: a 3-soluble one would need a faithful action of C_9 on a $\{2, 5, 11, 13\}$-subgroup of $\mathrm{GL}_5(3)$, and the only non-abelian simple group with cyclic Sylow 3-subgroups of order 9 whose order divides $|\mathrm{GL}_5(3)|$ is $\mathrm{PSL}_2(8)$, whose smallest faithful module

over $\mathbb{F}_3$ has dimension 7), so these six groups carry no fusion system. The pipeline now discards such candidates in Stage 1a.

References

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